GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

Interior and Exterior Angles of a Triangle

Question Bank 6.T2 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The third angle of a triangle with angles of 34° and 101°

45°

180 − 135.

2. The third angle of a right triangle with an angle of 27°

63°

180 − 90 − 27.

3. The exterior angle whose remote interior angles are 52° and 61°

113°

52 + 61.

4. The other remote interior angle, when an exterior angle is 118° and one remote angle is 47°

71°

118 − 47.

For questions 5 through 7, find x and the angles.

5. A triangle's angles are x°, $(x+30)^\circ$ and $(2x-10)^\circ$.

x = 40; 40°, 70° and 70°

4x + 20 = 180.

6. An isosceles triangle's base angles are each $(3x+5)^\circ$ and the third angle is 50°.

x = 20; 65° and 65°

6x + 10 + 50 = 180.

7. An exterior angle is $(6x-4)^\circ$; its remote interior angles are $(2x+8)^\circ$ and 40°.

x = 13; the exterior angle is 74°

6x − 4 = 2x + 48.

Choose one.

8. A triangle's angles are 30°, 70° and 80°. What is the exterior angle beside the 80°?

- ○ A. 80°
- ○ B. 280°
- ● **C. 100°**
- ○ D. 70°

C

30 + 70.

9. Which equation finds x when an exterior angle is $(4x)^\circ$ and its remote interior angles are 45° and $(x+15)^\circ$?

- ○ A. $4x+x+15=180$
- ● **B.** $4x=45+x+15$
- ○ C. $4x=x+15$
- ○ D. $4x+45+x+15=180$

B

Outside equals the two far ones: x = 20.